ref_checker:

What I built and how I built it

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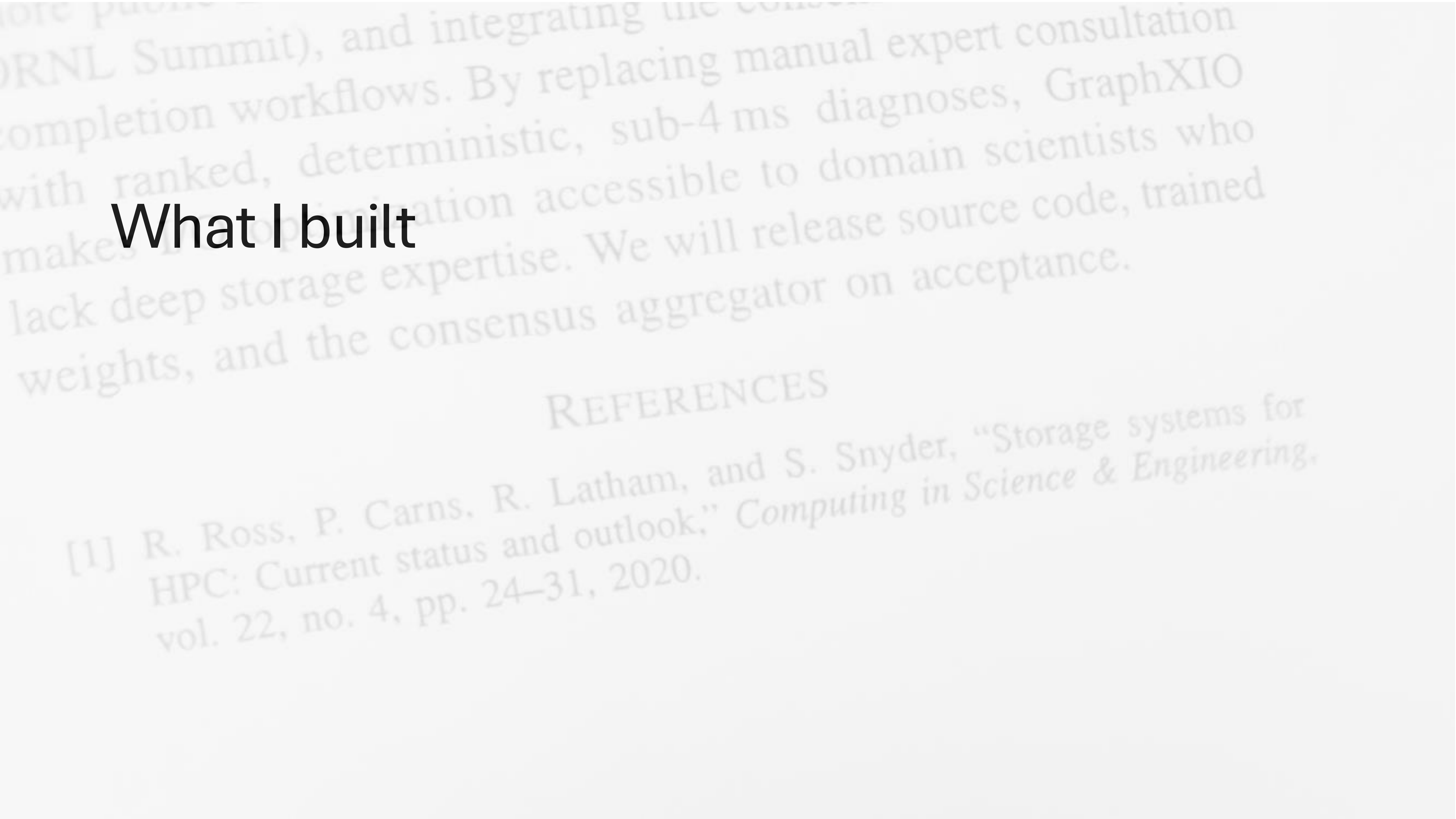
REFERENCES

[1] R. Ross, P. Carns, R. Latham, and S. Snyder, "Storage systems for HPC: Current status and outlook," *Computing in Science & Engineering*, vol. 22, no. 4, pp. 24–31, 2020.

What I built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REFERENCES
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ref_checker

https://github.com/rbross-hpc/ref-checker

Idea: CLI tool that tries "really hard" to find valid bibliographic information for all the references cited [in a paper].

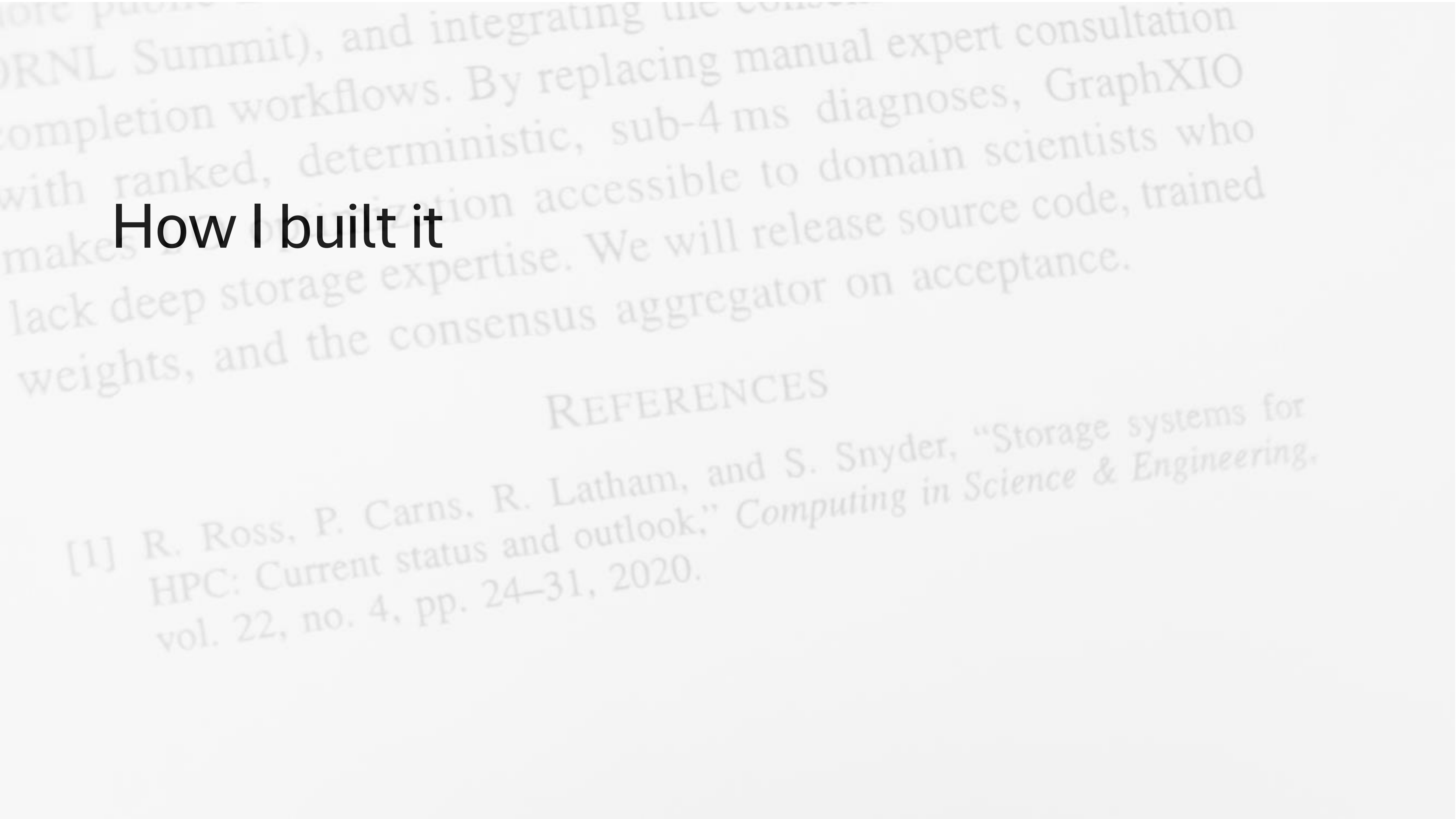
- Combination of heuristics and LLM (OpenAI API calls) to extract list of references.
- Support for looking up references in wide variety of (CS) sources: OpenAlex, CrossRef, DBLP, Semantic Scholar, arXiv
- Shortcuts when DOIs available
- Basic (not 404) check on non-paper URLs (e.g., GitHub repos)

```
(base) ➜  Downloads pipx install git+https://github.com/rbross-hpc/ref-checker.git
(base) ➜  Downloads ref-checker check pan_mcp-globus.pdf
< SNIP INFO OUTPUT >
[ref-checker] Extracted 54 reference(s).
[ref-checker] Refs cache: written to /Users/rross/Downloads/pan_mcp-globus.refs.json
[ref-checker] Checking 54 reference(s)...
[ref-checker] checking 1/54: 'Bio-Agents MCP: MCP servers for Protein Data Bank, ChEMBL, and other life science data'
[1] Chung, "Bio-Agents MCP: MCP servers for Protein Data Bank, ChEMBL, and other life science data", 2025
  OK (----)  https://github.com/dogeplusplus/bio-agents-mcp  [source: github]
[ref-checker] checking 2/54: 'Globus Platform Services for Data Publication'
[2] Ananthakrishnan et al., "Globus Platform Services for Data Publication", 2018, (Proceedings of the Practice and Experience on Advanced Research Computing: Seamless Creativity)
  OK (1.00)  doi:10.1145/3219104.3219127  [source: openalex]
[ref-checker] checking 3/54: 'Globus platform-as-a-service for collaborative science applications'
[3] Ananthakrishnan et al., "Globus platform-as-a-service for collaborative science applications", 2015, (Concurrency and Computation: Practice and Experience)
  OK (1.00)  doi:10.1002/cpe.3262  [source: openalex]
  Note: year mismatch (ref year=2015, match year=2014)
[ref-checker] checking 4/54: 'Model Context Protocol'
[4] "Model Context Protocol", 2024
  OK (----)  https://www.anthropic.com/news/model-context-protocol  [source: url]
  Note: URL liveness check only — no bibliographic record found
[ref-checker] checking 5/54: 'MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields'
[5] Batatia et al., "MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields", 2023
  OK (1.00)  doi:10.48550/arXiv.2206.07697  [source: arxiv]
  Note: year mismatch (ref year=2023, match year=2022)
< SNIP REMAINDER OF OUTPUT >
```

Misc.

- Full results of ref extraction and lookup in JSON output files
- Can also directly take a JSON list of references to check
- Caches results so you can re-run lookup
  (e.g., you’ve irked a service and it temporarily blocked you)
- Standard OpenAI environment variables for config
  (i.e., easy to point at Argo or a proxy)
- Stops looking when it finds a match
  (i.e., rather than hitting all services, goes faster)

How I built it



Environment

Kind of a defensive posture

- Linux desktop system on ANL network
  - No weird Argo proxy stuff needed
  - Running heavyweight stuff doesn’t impact my UI
- OpenCode running inside a Docker Container
  - It would have to deliberately jailbreak to wreck my machine
  - Opus 4.7 for planning, Sonnet 4.6 for building
- Tmux session with host and container windows
  - Attach/detach at will, resilient to network glitches
  - Easy to test things, push changes
- Shared workspace directory with git clone
  - Agent can make commits, but …
  - Container doesn’t have SSH keys, can’t push

Bonus: trick to avoid ProxyJump when you can

```
Host radix-llm
	User rross
	Hostname 140.221.15.15

Match originalhost radix-llm !exec "nc -nz -G 1 140.221.15.15 22”
	# was "nc -nz -w 1 140.221.15.15 22" (apple nc uses -G)
	ProxyJump rross@logins.cels.anl.gov
```

Q: What does that do?

A: This SSH config makes radix-llm an alias for the machine at 140.221.15.15, and conditionally uses a jump host only when the machine is not directly reachable.